Research Note: Polymarket Weather Markets — Selective Edge Trading

Author: ron1n97
Status: portfolio research write-up (curated extract from a private lab)
Stack: Python, Open-Meteo / station history, Polymarket CLOB weather buckets, paper + tiny-live loop
Code: this repository

Abstract

I studied Polymarket daily **max-temperature** bucket markets as a **mispricing** problem, not a "predict the exact °C" problem. Fair probability is compared to market-implied probability; trades fire only when edge clears filters for sample size, liquidity, and regime.

Headline backtest (selected filter `non_legacy_v2_mid`): 1,123 trades · hit-rate **52.5%** · avg **+0.026** PnL/share · total **+29.2** PnL/share units (2025-12-06 → 2026-03-22). The book is **NO-heavy**: `bet no` hit-rate **82.6%** / **+0.039**/share; `bet yes` hit-rate **9.7%** / **+0.008**/share.

Live status: paper + **$1** tiny-live on whitelist cities only. At note time the live journal had **4 open / 0 settled** trades — **not** evidence of live alpha. Bankroll curves below are **backtest simulations**, same caveat as in the BTC research note.

Working thesis — mispricing filter, not temperature oracle



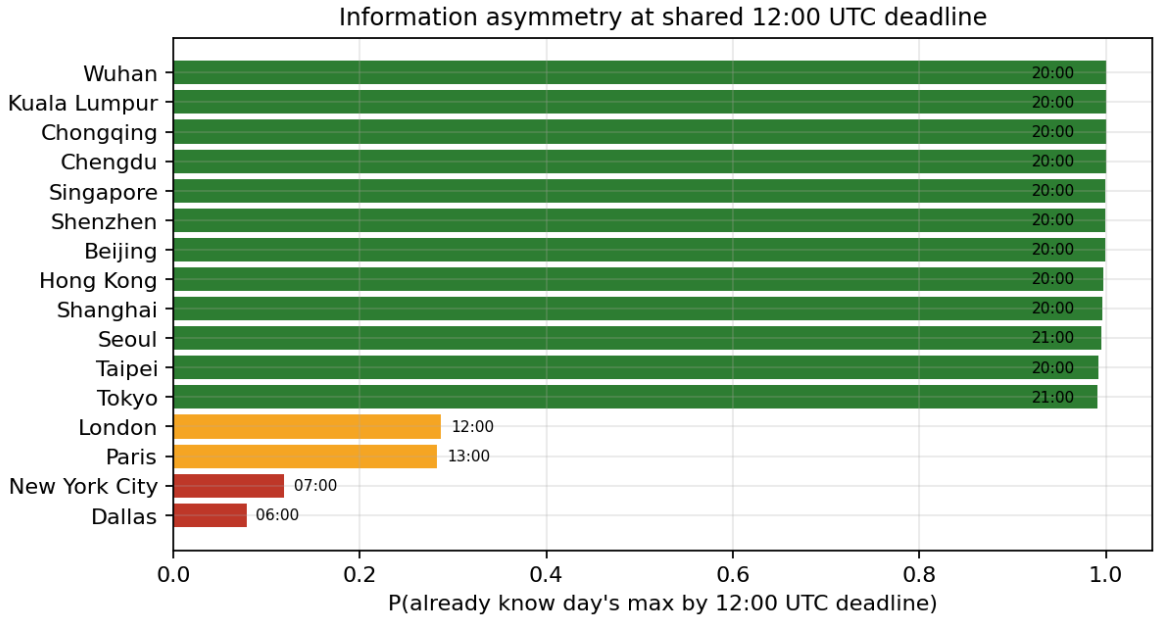
Research path

---

1. Market setup

- Product: multi-outcome daily high-temperature buckets (exact °C / tails)
- Shared trading deadline: **12:00 UTC** on resolution day
- Settlement: Weather Underground airport-station daily max (integer °C)
- Universe researched: up to ~39 cities; ~20–35 markets open on a typical day

Structural fact: the deadline is global, but **local information** at 12:00 UTC is not. In UTC+8/+9 cities the local day is already evening — today's max is often nearly known. In US West it is still night.



Deadline information asymmetry

That asymmetry is why **same-day Asia** sits at the top of the intended edge hierarchy — separate from day+4 forecast following.

---

2. Thesis

Do **not** mix these into one rule set:

Regime	Posture
Day+4	Low confidence; sparse edges; mostly hunt overpriced center → selective **NO**
Day+1 / Day+2	Better forecast quality; selective YES/NO
Same-day Asia (pre-12:00 UTC)	Strongest structural edge; use observed path, not just forecast

Decision unit = Polymarket's **actual buckets** (exact bin, 3-bin cluster, tail), with:

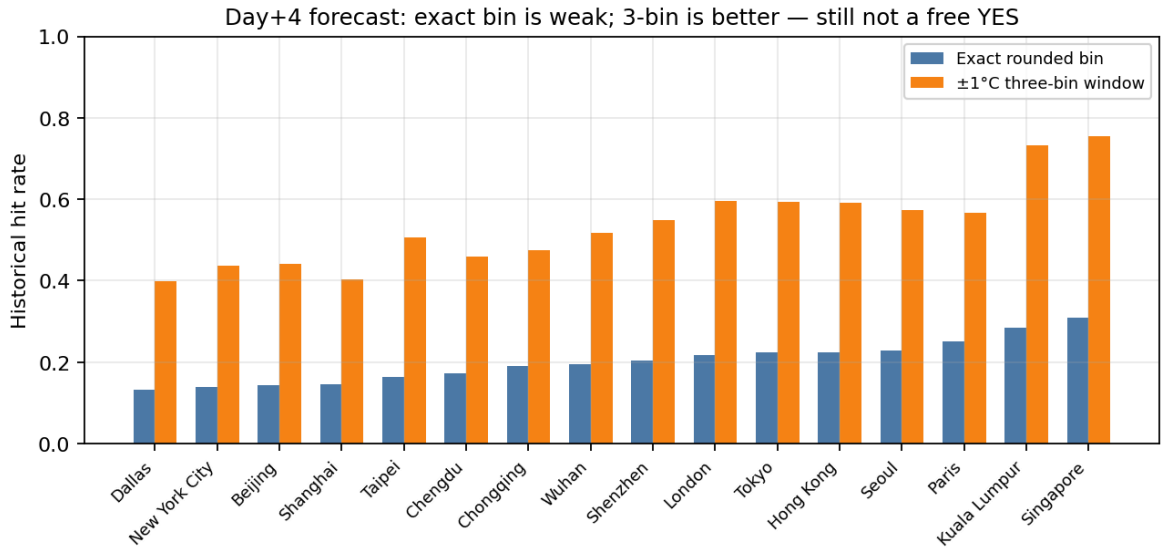
[= -]

Trade only when edge survives model-risk / spread / liquidity penalties.

---

3. Why naive forecast-following fails

Day+4 archived forecasts: **exact rounded bin** hit rates are low (often ~13–31% by city). A **±1°C three-bin** window is much higher (~40–75%), but that still does **not** justify blindly buying the center cluster at market prices — crowd already piles into the obvious bins.



Day+4 exact vs three-bin

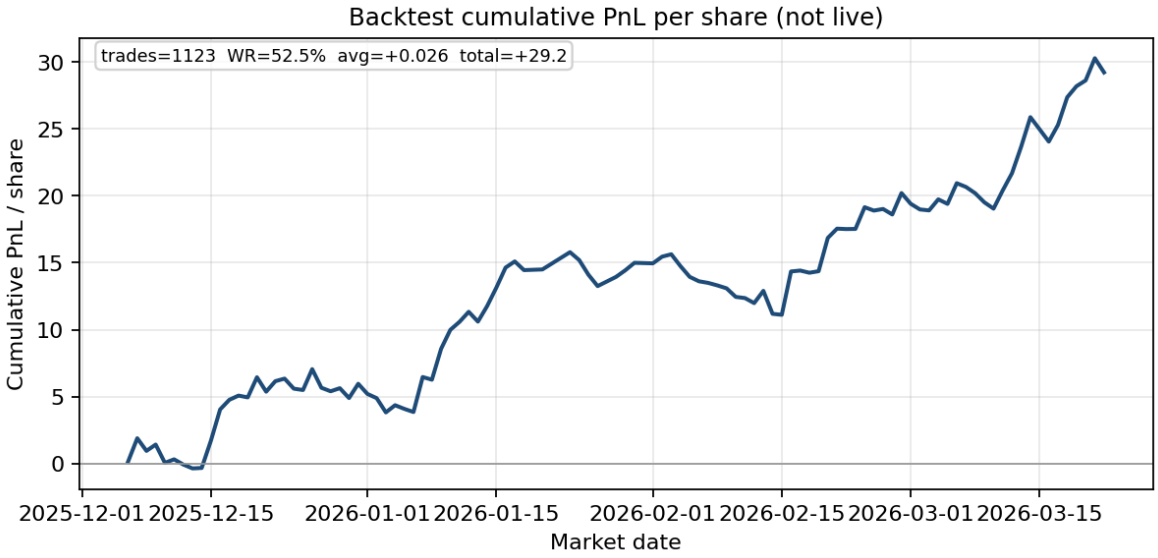
Practical implication: day+4 is **not** a broad long book; it may support selective **NO** / tail ideas when the center is overpriced.

---

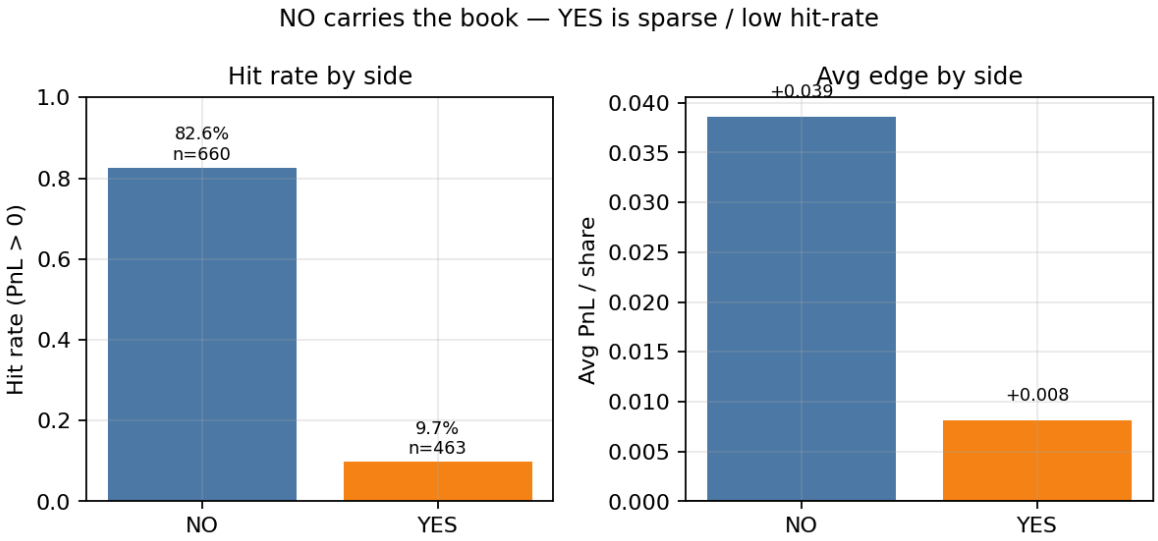
4. Selected edge backtest

Filter pack labeled **`non_legacy_v2_mid`** (lab naming): mid thresholds on edge / sample, non-legacy market set.

Metric	Value
Trades	**1,123**
Hit-rate (PnL > 0)	**52.5%**
Avg PnL / share	**+0.026**
Total PnL / share	**+29.2**
Date span	2025-12-06 → 2026-03-22
Side mix	660 NO / 463 YES



Cumulative backtest PnL

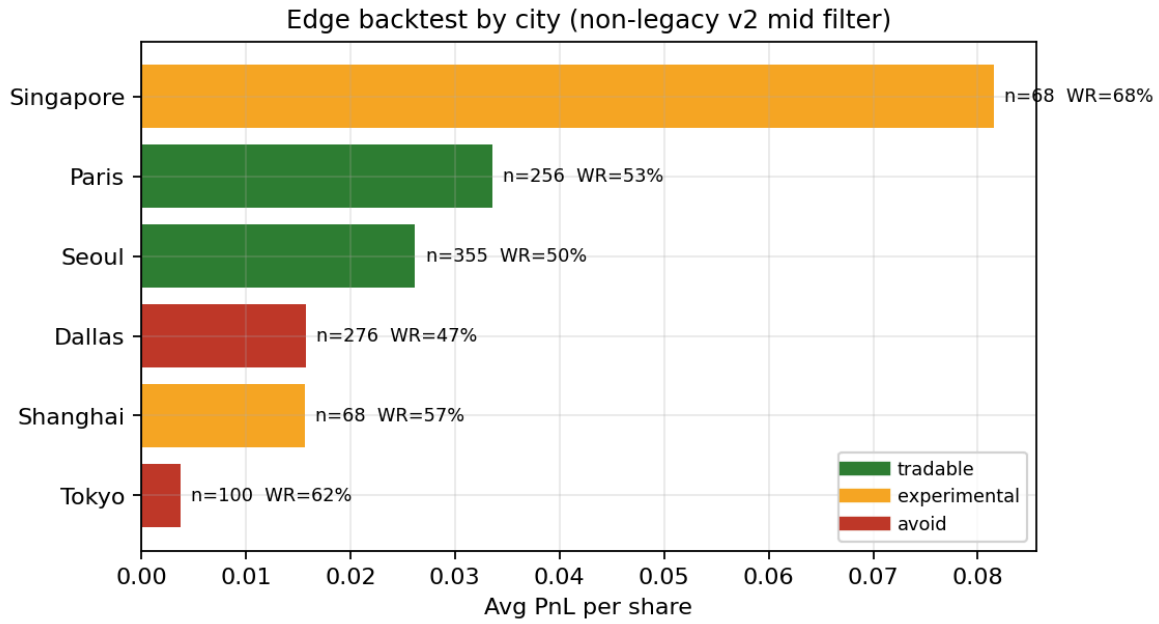


Side split

City view (same backtest) drives the live whitelist:

City	Tier	Trades	Avg PnL/share	Hit-rate
Singapore	experimental	68	+0.082	67.6%
Paris	**tradable**	256	+0.034	52.7%
Seoul	**tradable**	355	+0.026	49.9%
Dallas	avoid	276	+0.016	47.5%
Shanghai	experimental	68	+0.016	57.4%
Tokyo	avoid	100	+0.004	62.0%

Seoul/Paris earn **tradable** on sample + stability, not peak raw avg. Singapore looks hot but thin → experimental. Dallas/Tokyo deprioritized.



City avg PnL

---

5. Live policy (honest)

From `live_policy_v1.md` / `city_whitelist_v1.md`:

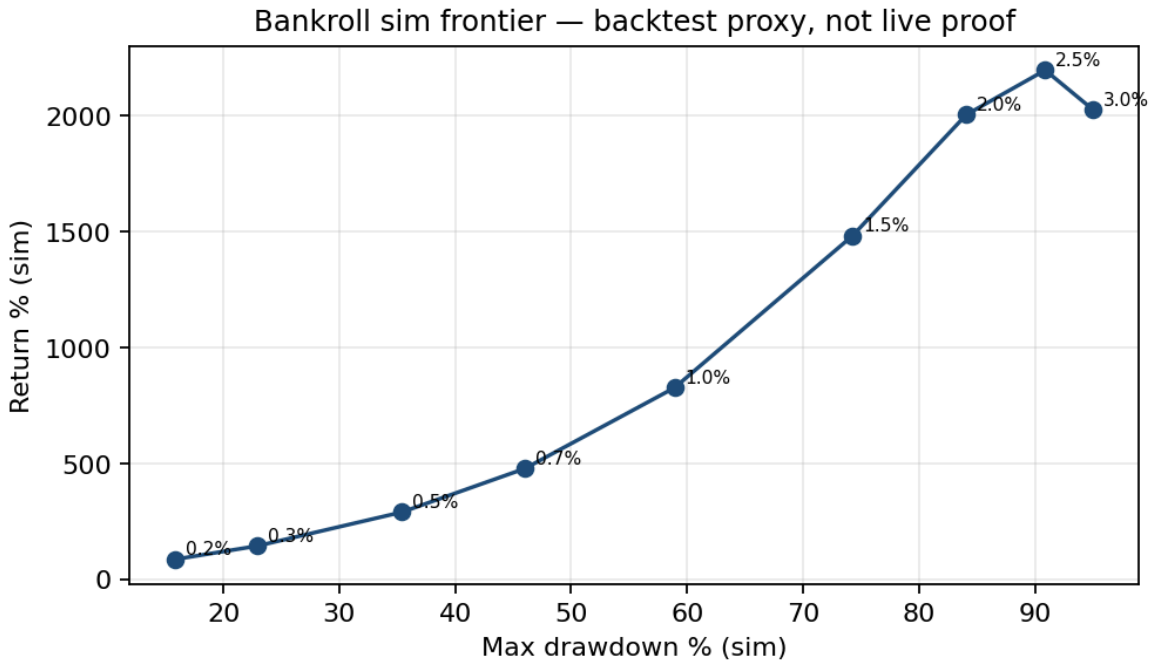
- Mode: **paper + tiny live** in parallel
- Live cities: **Seoul, Paris** only
- Size: **$1**/trade (Polymarket minimum binds at ~$50 bankroll ≈ 2%/trade)
- Default exit: hold to resolution
- Expand cities before size; need ~20–30 live trades before arguing live edge

At write-up time: **4 open, 0 settled, $0 realized live PnL**. Treat live as process validation, not performance.

---

6. Bankroll simulations (backtest only)

Fractional Kelly-style sweeps on the same 1,123-trade path (start $50) show the usual return / drawdown frontier. Useful for sizing intuition; **not** a claim about live compounding.



Bankroll frontier

Example: fraction 0.2% → ~+86% sim return / ~16% max DD; fraction 2% → large sim return with ~84% max DD. Live stays at $1.

---

7. What this repo is / is not

Is: curated research note, sanitized result tables, figures, strategy/policy docs, and the scripts that build the daily paper cycle / edge backtests.

Is not: a dump of raw `data/`, full `outputs/` archives, API keys, wallet state, or a claim of production alpha.

Private lab remains local; this is the public extract.

---

8. Takeaways

1. **Edge hierarchy ≠ forecast accuracy ranking.** Same-day Asia wins on information timing; day+4 needs selective NO, not center-bin YES spam.
2. **NO carries measured PnL** in the selected backtest; YES is low hit-rate and must stay sparse.
3. **Whitelist is an ops decision** (sample + stability), not "highest backtest city wins."
4. **Separate layers:** forecast eval ≠ edge backtest ≠ paper journal ≠ tiny live. Mixing them invents fake confidence.

Reproduce figures

```
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python scripts/make_figures.py
```

Full daily paper cycle needs local market/forecast fetches (see README). Result CSVs under `results/` are enough to regenerate the note figures.